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Wireless Networking

“Li-Fi Enables Business Analytics... Where HIGH-SPEED CONNECTIVITY And LIGHTING Play An IMPORTANT Role”

If you could get 100 times faster networks than traditional Wi-Fi, would you not invest in such futuristic connectivity? Anuj Dhir, vice president and business head - commercial lighting business, Wipro Enterprise Ltd explores growth opportunities of Li-Fi technology in India and Asia with Nimeesh Kumar of *Electronics For You*

Q. What are the business opportunities of Li-Fi in India and Asia?

A. Li-Fi is a high-speed, bi-directional, fully-networked wireless communication that uses visible light rather than radio frequencies. It offers significantly greater security, data speeds and densities to support more robust and reliable wireless networks that complement and enhance existing cellular and Wi-Fi networks.

Li-Fi is likely to open avenues for the service business model, namely, lighting as a service (LaaS), which bundles lighting and bandwidth.

Wipro Lighting has partnered with pureLiFi to harness their Li-Fi technology for developing applications for wireless communications and the lighting market in Asia.

Multinationals, especially IT companies, need unprecedented data and huge data bandwidths. Li-Fi enhances user experience with lights connecting to data, powering the industry's growing demand for connectivity.

Li-Fi networks for daily work, conference streaming, remote desktops along with video can provide a better user experience with robust security. Directionality of light propagation can effectively reduce interference in heavily populated offices. Wireless offloading to Li-Fi releases the spectrum for connecting other devices.

Q. Which sectors can see higher adaptability of Li-Fi?

A. Li-Fi offers an unprecedented opportunity for connectivity within hospitals and healthcare facilities. It

does not emit electromagnetic interference and, hence, does not interfere with medical instruments, including MRI scanners.

In defence and banking sectors, its applications include secure communications through light due to the intrinsically safe nature of containing light. It can also apply different security levels to individual lights or a group of lights, allowing for sophisticated geofencing.

Lights for streets, buildings and transportation will be able to communicate wirelessly. Li-Fi will relieve public wireless congestion as an off-loading facility for radio frequencies.

Soon, Li-Fi will also enable vehicle-to-vehicle communications. The technology is well-suited for aviation, too. Use of the visible light spectrum means it will not cause any electromagnetic interference with an airplane's crucial instruments.

Li-Fi enables business analytics across all the above-mentioned segments where high-speed connectivity and lighting play an important role.

Q. What is the best way to tap this opportunity?

A. Establishing a team of sales and marketing professionals who understand upcoming technologies and will cater to smart and connected lighting projects can go a long way. Also, exploring avenues where Li-Fi can be effectively put into place is important.

Q. Who is the target audience?

A. Progressive customers who are keen in the early adoption of new

technologies and trends are the key target customers.

Q. What is the role of partners here?

A. They can create awareness and engage interest in the market about Li-Fi.

Q. How can architects and other influencers in the ecosystem complement the technology?

A. Architects can help with design and aesthetics of the workspace. Whereas, consultants, especially Internet of Things (IoT) ones, can bring in new technologies to modern workspaces.

Q. As a leader in LED lighting solutions, would you like to say something to senior decision makers?

A. We must ensure that we are providing our customers with the gold standard in lighting innovation. Wipro envisions to provide high-speed, secure and safe data transmission using Li-Fi technology.

While technology upgrades, capex and opex limit the use of wired connectivity, limited radio frequency spectrum also puts constraints on increasing demand for ubiquitous connectivity with varied bandwidth capability. Moreover, use of the wireless spectrum loads the premium licensing cost of spectrum usage on telcos and total cost of ownership on end customers.

Li-Fi uses the unregulated spectrum of visible light (380nm to 750nm), which does not need licensing. Its adoption over time will bring down the total cost of ownership. **EFY**